

Arbeitsbericht

- Name: Denis Ermurachi
 - Datum: 28.09.2026
 - Laborübung: Standortvernetzung von 2 Standorten
 - Fach: SYTD
 - Klasse: 4AHITS
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1. Konfigurationen aller Geräte:

Device	Interface	IP Address / Prefix	Default Gateway
R-Br1	FE0/0	192.168.1.1 /24	
	FE0/1	10.0.0.1/30	
R-Sd1	FE0/0	192.168.2.1 /24	
	FE0/1	10.0.0.2/30	
S-Br1	VLAN 1	192.168.1.2 /24	192.168.1.1
S-Sd1	VLAN 1	192.168.2.2 /24	192.168.2.1
PC-A		192.168.1.111/24	192.168.1.1
PC-B		192.168.2.111/24	192.168.2.1

Am Router R-Br1 nu R-Sd1 werden noch statische Routen benötigt, um die Kommunikation zwischen den beiden Standorten zu ermöglichen.

R-Br1:

```
ip route 192.168.1.0 255.255.255.0 10.0.0.2
ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

R-Sd1:

```
ip route 192.168.2.0 255.255.255.0 10.0.0.1
ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

R-Br1

```
R-Br1#show ip interface brief
Interface          IP-Address      OK? Method Status        Protocol
FastEthernet0/0    192.168.1.1    YES manual up            up
FastEthernet0/1    10.0.0.1       YES manual up            up
Vlan1              unassigned     YES unset  administratively down down
```

R-Sd1

```
R-Sd1#show ip interfa
R-Sd1#show ip interface br
R-Sd1#show ip interface brief
Interface          IP-Address      OK? Method Status        Protocol
FastEthernet0/0    192.168.2.1    YES manual up            up
FastEthernet0/1    10.0.0.2       YES manual up            up
Vlan1              unassigned     YES unset  administratively down down
```

S-Br1

```
S-Br1#show ip in
S-Br1#show ip interface b
S-Br1#show ip interface brief
Interface          IP-Address      OK? Method Status        Protocol
FastEthernet0/1    unassigned     YES manual up            up
FastEthernet0/2    unassigned     YES manual up            up
FastEthernet0/3    unassigned     YES manual down          down
FastEthernet0/4    unassigned     YES manual down          down
FastEthernet0/5    unassigned     YES manual down          down
FastEthernet0/6    unassigned     YES manual down          down
FastEthernet0/7    unassigned     YES manual down          down
FastEthernet0/8    unassigned     YES manual down          down
FastEthernet0/9    unassigned     YES manual down          down
FastEthernet0/10   unassigned     YES manual down          down
FastEthernet0/11   unassigned     YES manual down          down
FastEthernet0/12   unassigned     YES manual down          down
FastEthernet0/13   unassigned     YES manual down          down
FastEthernet0/14   unassigned     YES manual down          down
FastEthernet0/15   unassigned     YES manual down          down
FastEthernet0/16   unassigned     YES manual down          down
FastEthernet0/17   unassigned     YES manual down          down
FastEthernet0/18   unassigned     YES manual down          down
FastEthernet0/19   unassigned     YES manual down          down
FastEthernet0/20   unassigned     YES manual down          down
FastEthernet0/21   unassigned     YES manual down          down
FastEthernet0/22   unassigned     YES manual down          down
FastEthernet0/23   unassigned     YES manual down          down
FastEthernet0/24   unassigned     YES manual down          down
GigabitEthernet0/1 unassigned     YES manual down          down
GigabitEthernet0/2 unassigned     YES manual down          down
Vlan1              192.168.1.2    YES manual up            up
```

S-Br2

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up
FastEthernet0/3	unassigned	YES	manual	down	down
FastEthernet0/4	unassigned	YES	manual	down	down
FastEthernet0/5	unassigned	YES	manual	down	down
FastEthernet0/6	unassigned	YES	manual	down	down
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	down	down
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down
FastEthernet0/22	unassigned	YES	manual	down	down
FastEthernet0/23	unassigned	YES	manual	down	down
FastEthernet0/24	unassigned	YES	manual	down	down
GigabitEthernet0/1	unassigned	YES	manual	down	down
GigabitEthernet0/2	unassigned	YES	manual	down	down
Vlan1	192.168.2.2	YES	manual	up	up

2. Überprüfen Sie die End-to-End-Konnektivität:

PC-A zu S1/R1:

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=6ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms
```

Mit einem ping auf R1 FE0/0 Schnittstelle wurde die Konnektivität des Switches überprüft.

PC-A zu R2/S2:

```
C:\>tracert 192.168.2.2

Tracing route to 192.168.2.2 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.1.1
  2  0 ms    0 ms    0 ms    10.0.0.2
  3  0 ms    0 ms    0 ms    192.168.2.2

Trace complete.
```

Mit einem tracert auf R2 FE0/0 Schnittstelle wurde die Konnektivität des R-Br1 und R-Sd1 überprüft.

PC-A zu PC-B:

```
C:\>ping 192.168.2.111

Pinging 192.168.2.111 with 32 bytes of data:

Reply from 192.168.2.111: bytes=32 time<1ms TTL=126
Reply from 192.168.2.111: bytes=32 time<1ms TTL=126
Reply from 192.168.2.111: bytes=32 time<1ms TTL=126
Reply from 192.168.2.111: bytes=32 time=8ms TTL=126

Ping statistics for 192.168.2.111:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms
```

Mit einem ping auf PC-B wurde die End-to-End-Konnektivität aller Geräte überprüft.

```
C:\>tracert 192.168.2.111

Tracing route to 192.168.2.111 over a maximum of 30 hops:

  1  0 ms    0 ms    1 ms    192.168.1.1
  2  0 ms    0 ms    0 ms    10.0.0.2
  3  0 ms    0 ms    0 ms    192.168.2.111
```

Der tracert auf PC-B zeigt den vollständigen Pfad der Pakete von PC-A zu PC-B und bestätigt, dass die Routing-Konfigurationen korrekt sind und die End-to-End-Konnektivität funktioniert.

PC-B zu PC-A:

```
C:\>tracert 192.168.1.111

Tracing route to 192.168.1.111 over a maximum of 30 hops:

  1  0 ms    3 ms    0 ms    192.168.2.1
  2  0 ms    0 ms    0 ms    10.0.0.1
  3  0 ms    0 ms    0 ms    192.168.1.111

Trace complete.

C:\>
```

```
C:\>ping 192.168.1.111

Pinging 192.168.1.111 with 32 bytes of data:

Reply from 192.168.1.111: bytes=32 time<1ms TTL=126
Reply from 192.168.1.111: bytes=32 time<1ms TTL=126
Reply from 192.168.1.111: bytes=32 time<1ms TTL=126
Reply from 192.168.1.111: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.111:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

ping und tracert von PC-B zu PC-A sind ebenfalls erfolgreich.